

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Module - I

Practical – VI

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Class:TYIT	Batch:2
Date of Assignment: 21/08/26	Date/Time of Submission: 21/08/26

6.Designing Effective and Visually Appealing Dashboards.

a)Apply worksheet-level and workbook-level formatting.

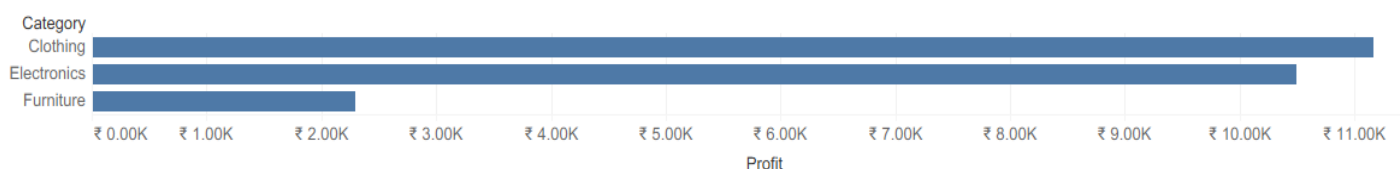
STEPS:-

1. Connect List of Orders.csv and Order Details.csv.
2. Create a relationship using Order ID.
3. Create Category Profitability using Category and Profit.
4. Apply consistent fonts and title formatting.
5. Remove unnecessary grid lines and borders.

OUTPUT:-

The screenshot displays the Tableau interface. On the left, the 'Connections' pane shows 'List of Orders' as a text file. The main workspace shows a relationship between 'List of Orders.csv' and 'Order Details.csv' using the 'Order ID' field. The 'Format Lines' dialog is open, showing options for 'Default', 'Total', and 'Grand Total' formatting, including font style (Regular, 10pt), font size (Seogoe UI, 1..), and color selection. The 'Format Lines' dialog also shows options for 'Lines', 'Zero Lines', 'Trend Lines', 'Ref Lines', 'Drop Lines', 'Axis Rulers', and 'Axis Ticks'.

Category Profitability T054



INSIGHT:

Consistent formatting makes the dashboard clean, professional, and easy to read.

b) Customize number, date, and null value formatting.

STEPS:-

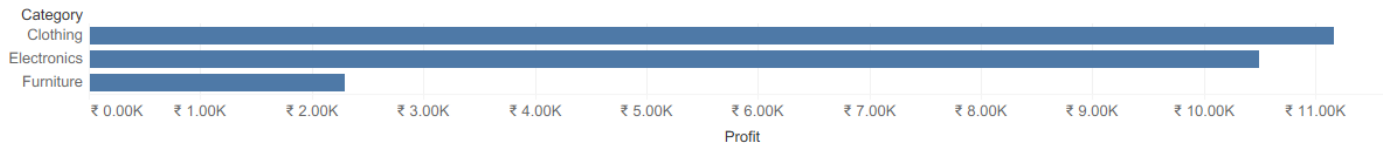
1. Format Profit and Amount as Indian Currency ₹.
2. Set decimal places to 0.
3. Format Order Date as dd-MMM-yyyy.
4. Check for NULL values.
5. Display NULL values as N/A or Not Reported.

OUTPUT:-

The screenshot shows a data visualization tool interface. On the left, a tree view lists data sources: 'City', 'List of Orders.csv (Cou...', and 'Order Details.csv'. Under 'Order Details.csv', there are fields: 'Category', 'Order ID (Order Details..', 'Sub-Category', 'Amount', 'Profit', 'Quantity', and 'Order Details.csv (Cou...'. A context menu is open over 'Order Details.csv', showing options like 'Default Properties', 'Geographic Role', 'Group by', 'Folders', 'Replace References...', and 'Describe...'. The 'Number Format...' option is selected, opening a 'Default Number Format [Profit]' dialog box. The dialog box has a list of formats on the left: 'Automatic', 'Number (Standard)', 'Number (Custom)', 'Currency (Standard)', 'Currency (Custom)' (selected), 'Scientific', 'Percentage', and 'Custom'. On the right, settings for 'Currency (Custom)' are shown: 'Decimal places' is 2, 'Negative values' is '₹ -1234', 'Display Units' is 'Thousands (K)', 'Prefix / Suffix' is '₹', and 'Include thousands separators' is checked. Below the dialog box, a table is visible with columns: 'Customer Name', 'Order Date', 'State', 'City', 'Amount', 'Profit', 'Quantity', and 'Order Details.csv (Cou...'. The table has two rows of data: 'Electronics' and 'Furniture', both with a profit of '₹ 0.00K'. A context menu is also open over the 'Order Date' column, showing options like 'Add to Sheet', 'Show Filter', 'Duplicate', 'Rename', 'Hide', 'Create', 'Transform', 'Convert to Continuous', 'Change Data Type', 'Default Properties', 'Group by', 'Folders', and 'Hierarchy'. The 'Date Format...' option is selected, opening a 'Default Date Format [Order Date]' dialog box.

The 'Default Date Format [Order Date]' dialog box is shown. It has a list of date formats on the left: 'Automatic', 'Standard Long Date', 'Standard Short Date', '14-03-2001' (selected), '14-03-01', '14-3-01', '14.3.01', '2001-03-14', '3/14/2001', '3/14/01', '03/14/01', '03/14/2001', '14 March 2001', '14 March 2001', 'Wednesday, 14 March, 2001', 'Wed, March 14, 2001', and 'March 14, 2001'. The right side of the dialog is empty. At the bottom, there are 'Clear', 'OK', and 'Cancel' buttons.

Category Profitability T054



INSIGHT:

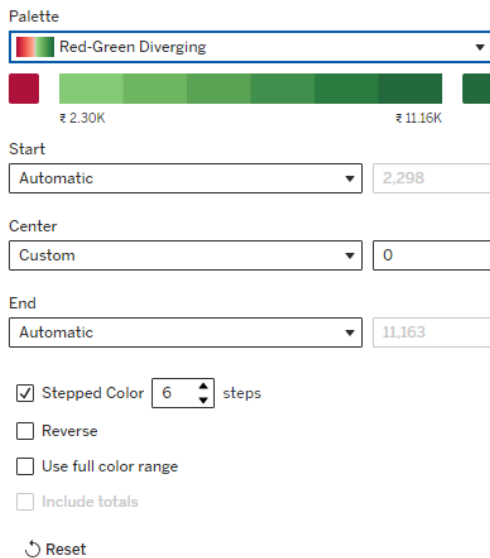
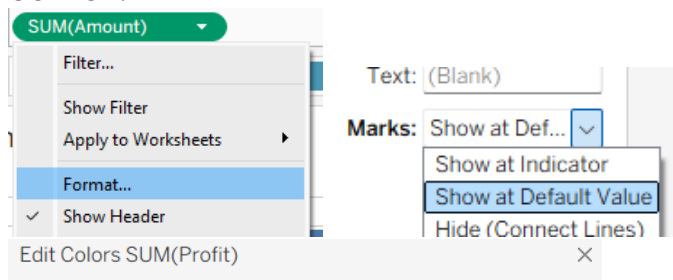
Proper formatting makes currency, dates, and missing values easier to understand.

c) Design effective color schemes and layouts.

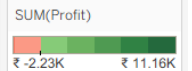
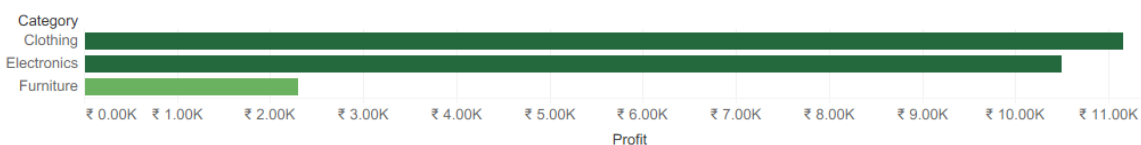
STEPS:-

1. Drag Profit to Color.
2. Apply a Diverging Color Scheme.
3. Set the color center to 0.
4. Create KPI Summary for:
Total Sales ○ Total Profit
○ Total Orders
5. Create the Executive Sales Overview dashboard.
6. Arrange the sheets neatly.

OUTPUT:-



Category Profitability T054



INSIGHT:

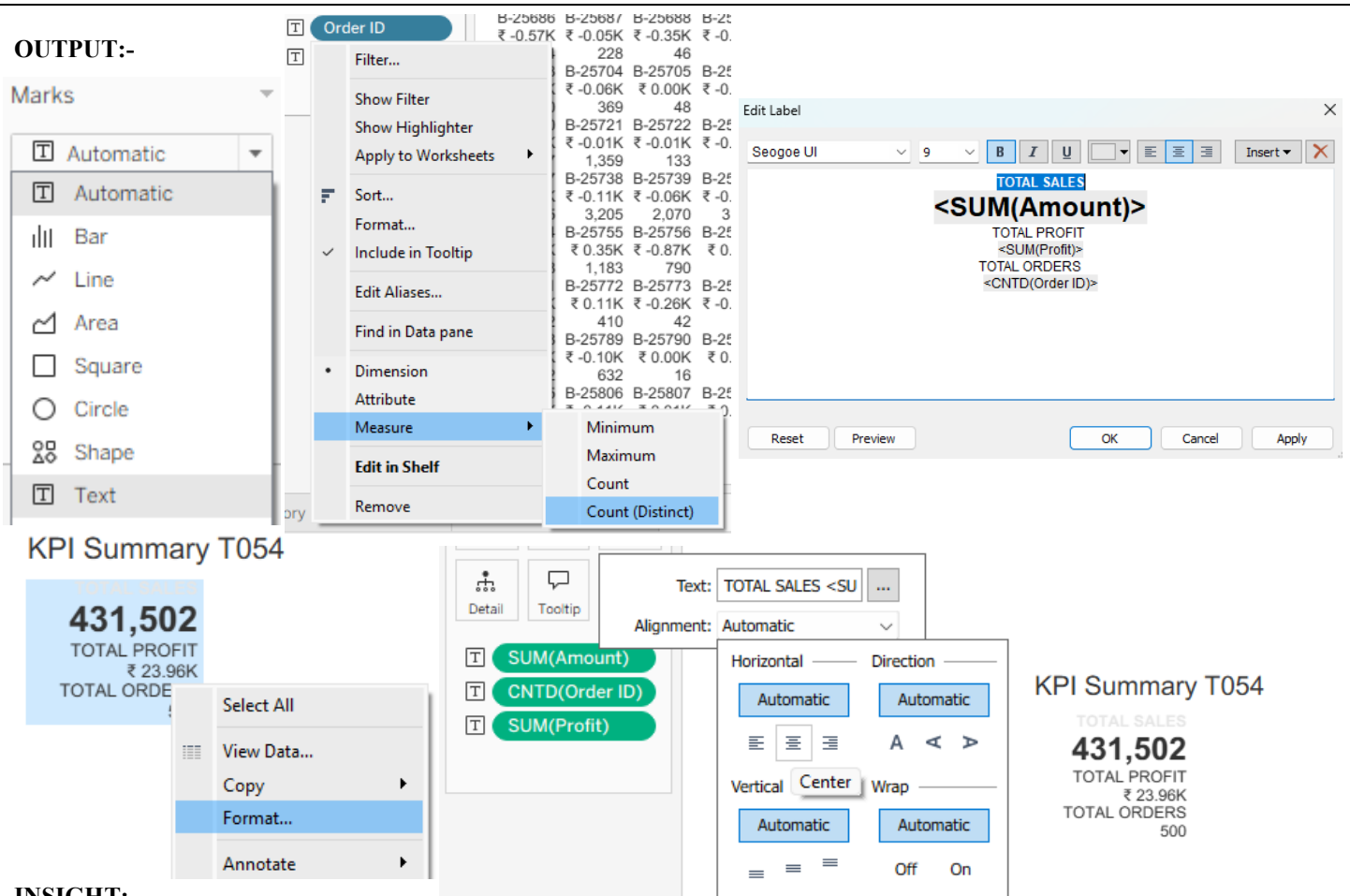
Colors quickly identify profit and loss, while KPI cards provide a quick business summary.

d) Create informative tooltips.

STEPS:-

1. Open Category Profitability.
2. Click Tooltip on the Marks card.
3. Add Amount, Quantity, and Order ID.
4. Change Order ID to Count Distinct.
5. Create a short sentence-based tooltip.
6. Test it by hovering over the bars.

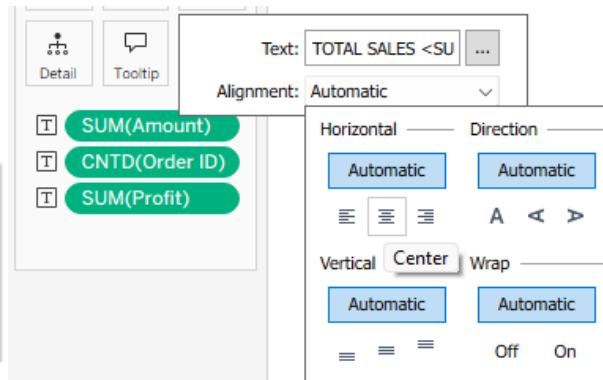
OUTPUT:-



KPI Summary T054

431,502
TOTAL PROFIT
₹ 23.96K
TOTAL ORDERS
500

- Select All
- View Data...
- Copy
- Format...
- Annotate



KPI Summary T054

TOTAL SALES
431,502
TOTAL PROFIT
₹ 23.96K
TOTAL ORDERS
500

INSIGHT:

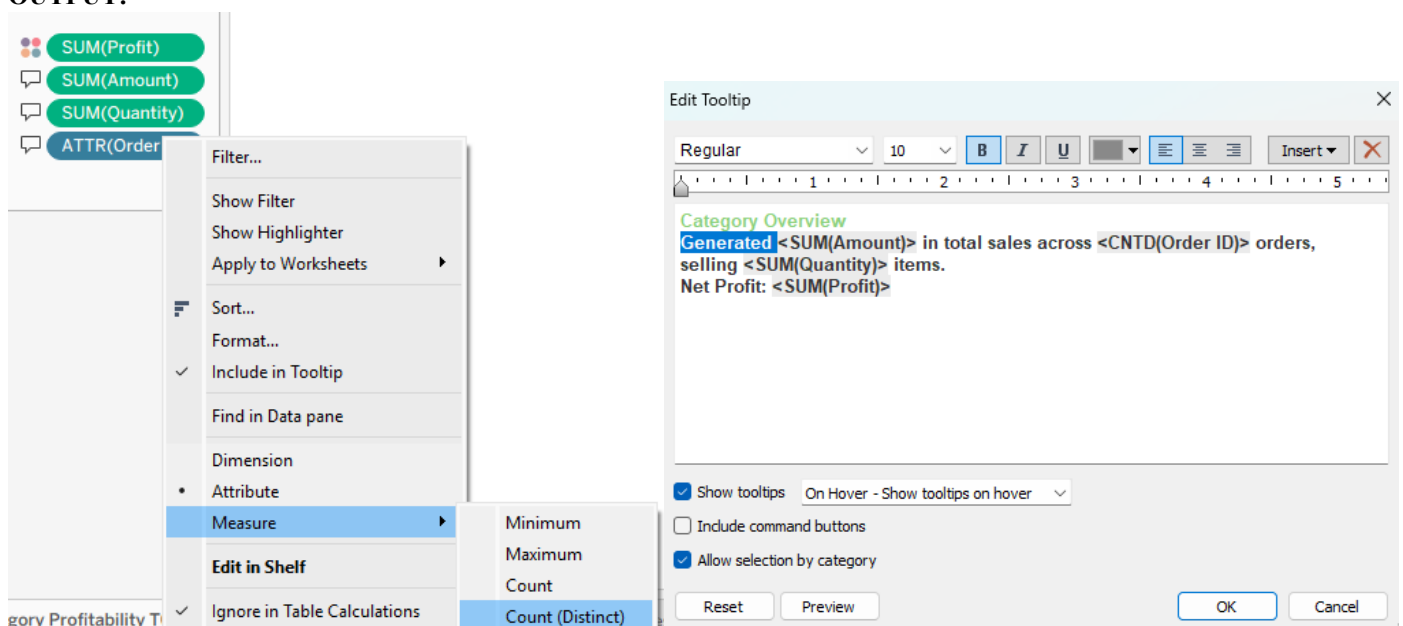
Custom tooltips provide extra information without adding another chart.

e) Implement "Viz in Tooltip".

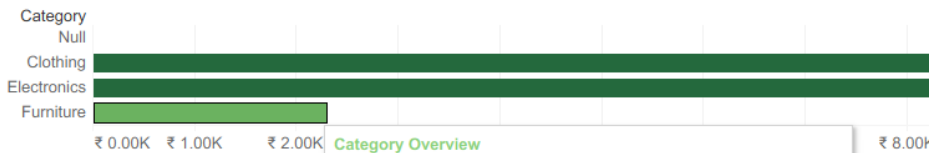
STEPS:-

1. Create a new sheet named Tooltip SubCategory Detail.
2. Add Sub-Category to Rows and Profit to Columns.
3. Apply color and sort by Profit.
4. Open the main chart's Tooltip.
5. Insert the detail sheet using Insert → Sheets.
6. Hover over different categories to test it.

OUTPUT:-



Category Profitability T054



Category Overview
Generated 127,181 in total sales across <CNTD(Order ID)> orders,
selling 945 items.
Net Profit: ₹ 2.30K

Sub-Category

Filter...

Show Filter

Show Highlighter

Apply to Worksheets

Sort...

Clear Sort

Standard

Standard

Fit Width

Fit Height

Entire View

Sort [Sub-Category]

Sort By

Data source order

Sort Order

☐ Ascending

☒ Descending

Clear

Edit Tooltip

Regular

10

B

I

U

Insert

Category Overview

Generated <SUM(Amount)> in total sales across <CNTD(Order ID)> o
selling <SUM(Quantity)> items.
Net Profit: <SUM(Profit)>

Sheets

City Sales And Nulls T054

Data Source Name

Sheet 1

Data Update Time

Sheet 4

Sheet Name

Sheet 5

Workbook Name

Page Count

Edit Tooltip

Regular

10

B

I

U

Insert

Category Overview

Generated <SUM(Amount)> in total sales across <CNTD(Order ID)> orders,
selling <SUM(Quantity)> items.
Net Profit: <SUM(Profit)>

<Sheet name="Sheet 5" maxwidth="300" maxheight="300" filter="<All Fields">

☒ Show tooltips

On Hover - Show tooltips on hover

☐ Include command buttons

☒ Allow selection by category

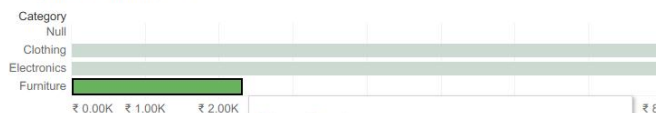
Reset

Preview

OK

Cancel

Category Profitability T054



Category Overview
Generated 127,181 in total sales across <CNTD(Order ID)> orders,
selling 945 items.
Net Profit: ₹ 2.30K

Sub-Category

Tables

Furnishings

Chairs

Bookcases

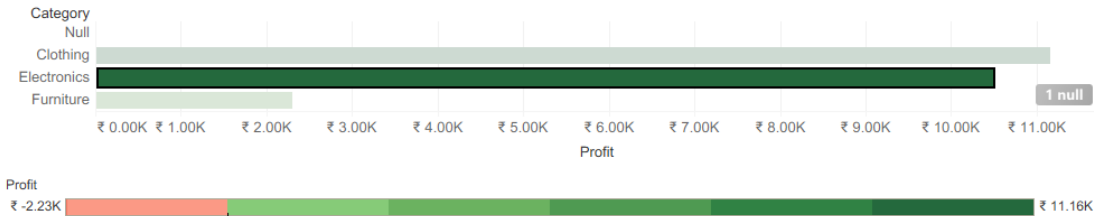
₹ 0.00K ₹ 5.00K

Profit

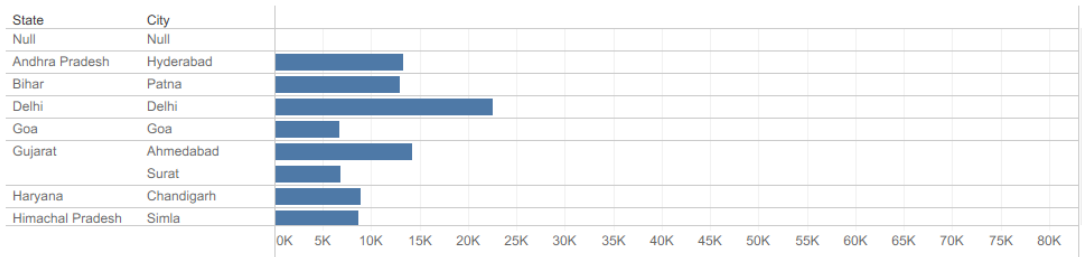
KPI Summary T054

TOTAL SALES
431,502
TOTAL PROFIT
₹ 23.96K
TOTAL ORDERS
500

Category Profitability T054



City Sales And Nulls T054



INSIGHT:

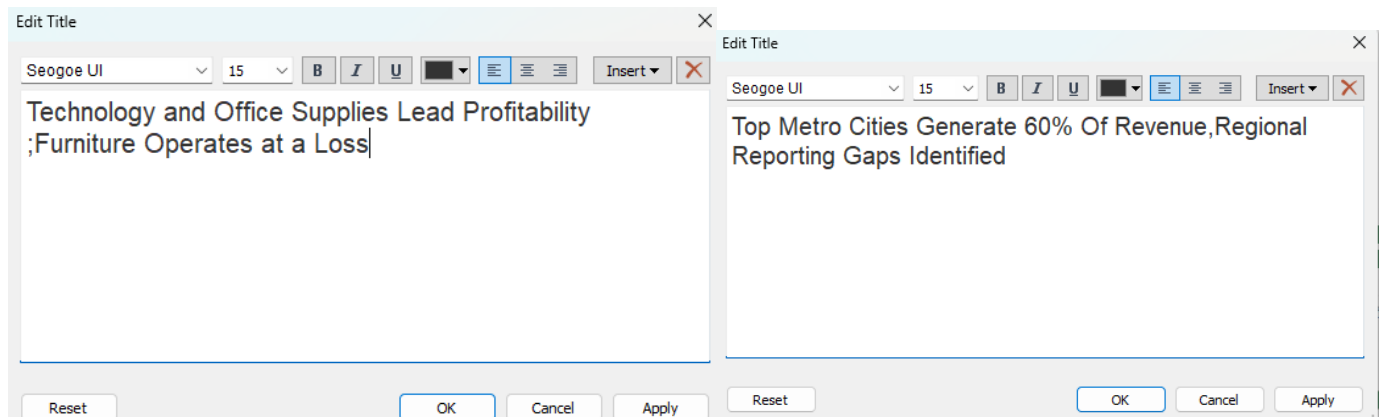
Viz in Tooltip shows detailed sub-category performance without making the dashboard crowded.

f) Improve readability and user experience.

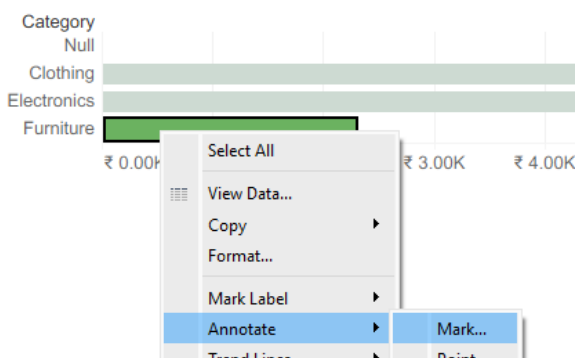
STEPS:-

1. Add equal padding around dashboard components.
2. Go to Dashboard → Actions.
3. Add a Highlight Action.
4. Set it to work On Hover using Category.
5. Remove unnecessary legends.
6. Check the dashboard in Presentation Mode.

OUTPUT:-



Technology and Office Supplies Lead Profitability



Edit Annotation

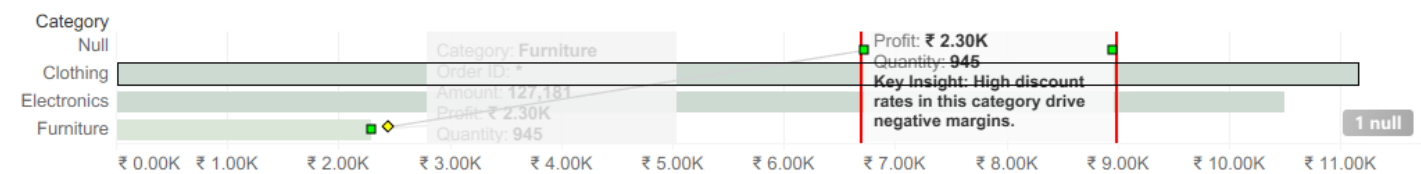
Seogoe UI 10 B I U [Color Picker] [List Icon] [List Icon] [List Icon] Insert X

Category: <Category>
 Order ID: <ATTR(Order ID)>
 Amount: <SUM(Amount)>
 Profit: <SUM(Profit)>
 Quantity: <SUM(Quantity)>

Key Insight: High discount rates in this category drive negative margins.

OK Cancel

Technology and Office Supplies Lead Profitability ;Furniture Operates at a Loss



INSIGHT:

Proper spacing, highlighting, and clean legends make the dashboard easier to use and understand.

g) Apply best practices for visual storytelling.

STEPS:-

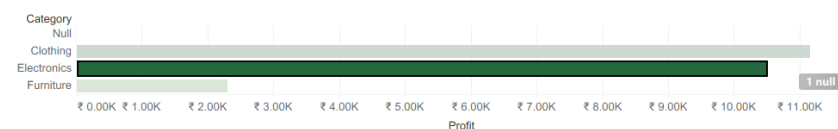
1. Arrange the dashboard using the F-pattern.
2. Keep KPI Summary at the top-left.
3. Use insight-based titles for the charts.
4. Add an annotation to an important or low-profit category.
5. Create Tableau Story Points to present the main findings.
6. Save and check the final dashboard.

OUTPUT:-

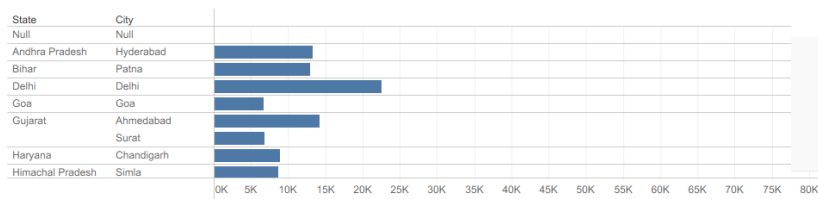
KPI
Summary
T054

TOTAL SALES
431,502
TOTAL PROFIT
₹ 23.99K
TOTAL ORDERS
500

Category Profitability T054



City Sales And Nulls T054



Downloads NetSim
 Documents Earlier this week
 Pictures SY20

File name: Lesson 6 video
 Save as type: Tableau Workbook (*.twb)

<

Executive Summary:Sales remain Strong across top categoriesreaching target revenue

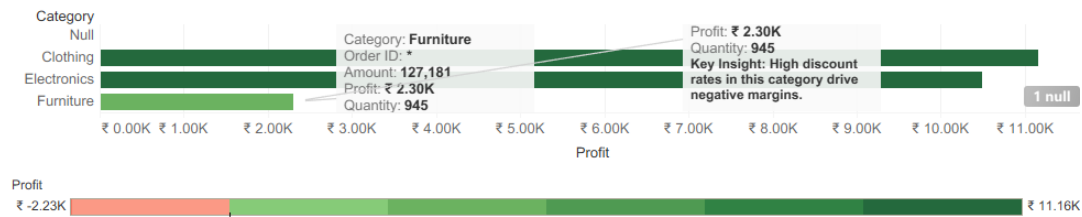
Profit Bottleneck Furniture margins are dragged down by shipping costs and return rates

>

KPI
Summary
T054

TOTAL SALES
431,502
TOTAL PROFIT
₹ 23.96K
TOTAL ORDERS
500

Technology and Office Supplies Lead Profitability ;Furniture Operates at a Loss



Top Metro Cities Generate 60% Of Revenue,Regional Reporting Gaps Identified

INSIGHT:

A good layout, meaningful titles, annotations, and Story Points make the main business insights easy to understand.

CONCLUSION:

Thus, this practical helped in understanding how to design effective and visually appealing dashboards by applying proper worksheet and workbook formatting, customizing number, date, and null values, and selecting suitable colors and layouts. The use of informative tooltips and Viz in Tooltip enhanced the presentation and made the dashboard more interactive. Overall, these techniques improved readability, user experience, and visual storytelling, making the dashboard clear, attractive, and easier to understand for effective data analysis.